NAIC 2026 · PROBLEM STATEMENT 01 · DEVELOPER INFRASTRUCTURE

natlas-health

Technical documentation: the open-source developer kit for building with N-ATLaS

Components	Python SDK · browser playground · 5-language evaluation suite · fine-tuning starter kit · English and Pidgin docs
N-ATLaS integration	NCAIR1/N-ATLaS (chat), NCAIR Hausa/Igbo/Yoruba/English ASR, gateway TTS voices, via N-ATLAS-Kit
Languages	English, Hausa, Yoruba, Igbo, Nigerian Pidgin
Repository	github.com/onifade617/N-ATLAS-enhanced-apps
Licence	Apache-2.0
Date	12 October 2026

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1. Developer guide: SDK, playground, evaluation, CLI

Source: docs/sdk/README.md

English · Naijá (Pidgin)

`natlas-health` helps developers build health applications on N-ATLaS in English, Hausa, Yoruba, Igbo and Nigerian Pidgin. It talks directly to the N-ATLaS gateway (N-ATLAS-Kit) for chat, speech-to-text and voices. It wraps no other model.

Part	What it gives you	Where
Python SDK	`NatlasClient`: typed errors, retries for cold starts, per-language ASR/TTS mapping. Standard library only.	`natlas_health/client.py`
Health safety layer	`HealthAssistant`: grounded prompts, multilingual danger-sign detection, emergency notice, template fallback	`natlas_health/assistant.py`, `safety.py`, `prompts.py`
Browser playground	Try chat, grounded answers, recording/transcription and voices. Shows the exact request, the checks and the Python code.	`natlas-health playground`
Evaluation	20-case health suite in 5 languages: safety, grounding, format and language checks, Markdown/JSON reports, CI pass bar	`natlas-health eval`
Fine-tuning starter kit	Build, validate and split chat data; LoRA/QLoRA script; 200-example seed set	`natlas-health dataset`, `finetune/`
Testing tools	`MockClient` (offline) and `FakeGateway` (local HTTP), so tests and CI never spend GPU credit	`natlas_health/testing.py`
Reference app	Lafiya AI, a full Django health platform built on this kit	repository root

Install

```
git clone <this repo> && cd <repo>
pip install -e .                 # SDK + CLI, no dependencies
pip install -e ".[audio]"        # + WAV→MP3 for WhatsApp voice notes
pip install -e ".[finetune]"     # + torch/transformers/peft/trl for fine-tuning
```

Python 3.9 or newer.

Connect to N-ATLaS

Deploy the N-ATLAS-Kit gateway (Modal or a local 24 GB GPU; see the self-hosting guide). Then set:

```
NATLAS_BASE_URL=https://<workspace>--natlas-serve-natlasservice-serve.modal.run
NATLAS_API_KEY=<one of the gateway's NATLAS_API_KEYS - not your Hugging Face token>
NATLAS_MODEL=NCAIR1/N-ATLaS      # optional
NATLAS_TIMEOUT=60                # optional, seconds
```

The CLI also reads these values from a `.env` file in the current directory. Check the connection:

```
natlas-health health
natlas-health chat "Sannu! Yaya kake?" --language ha
```

No GPU yet? Add `--mock` to any command to use the offline `MockClient`.

SDK in 2 minutes

```
from natlas_health import NatlasClient, NatlasError

client = NatlasClient.from_env(retries=2)          # retries cover a Modal cold start

reply = client.chat("Ẹ kú àárọ̀! Báwo ni?", language="yo")
print(reply.text, reply.latency_ms)

text = client.transcribe("voice_note.ogg", language="ha").text    # Hausa Whisper model
speech = client.speak("Sannu da zuwa.", language="ha")            # WAV bytes from an N-ATLaS voice
open("hello.wav", "wb").write(speech.audio)
```

Method	Gateway route	Returns
`health(timeout=10)`	`GET /health`	`HealthStatus(ok, details)`. It never raises.
`chat(messages, language=None, temperature=0.3, max_tokens=500)`	`POST /v1/chat/completions`	`ChatResponse(text, model, latency_ms, raw)`
`transcribe(audio, filename, content_type, language)`	`POST /v1/audio/transcriptions`	`Transcription(text, language, latency_ms, raw)`
`speak(text, language)`	`POST /v1/audio/speech`	`Speech(audio, content_type, voice, latency_ms)`

`messages` can be a plain string or an OpenAI-style list. `audio` can be bytes or a file path.

Languages. Use `en`, `ha`, `yo`, `ig` or `pcm`. Pidgin has no dedicated chat template or ASR model, so the SDK leaves `language` out of chat calls for Pidgin and uses the Nigerian-accented English ASR model and English voice. You don't need to handle this yourself.

Errors. Every failure is a `NatlasError` subclass, so you can handle each case or catch them all at once:

Error	When
`NotConfiguredError`	no base URL
`AuthenticationError`	401/403: wrong or missing `NATLAS_API_KEY`
`UnavailableError`	gateway down, cold-starting, timed out, 429/502/503/504. Retried if `retries > 0`
`FeatureNotEnabledError`	501, e.g. the gateway has speech switched off
`BadRequestError`	other 4xx (empty input, file too large)
`InvalidResponseError`	2xx with a body that doesn't match the API

Safe, grounded health answers

Health apps must not let a language model make up facts or miss an emergency. `HealthAssistant` uses a simple pattern:

1. **Your app** collects FACTS from the person's own records and GUIDANCE from vetted sources (WHO, NPHCDA).
2. **N-ATLaS** only puts them into words, in the person's language. The system prompt forbids diagnosing, giving doses, inventing numbers or addresses, and using markdown.
3. **The SDK** checks the question for danger signs with transparent keyword lists. If it finds one, the reply always starts with a fixed emergency notice ("go now / call 112"), whatever the model wrote. If N-ATLaS is unavailable, you get your `fallback` text instead of an exception.

```
from natlas_health import HealthAssistant, NatlasClient

assistant = HealthAssistant(NatlasClient.from_env(), assistant_name="MamaCare")
answer = assistant.answer(
    "Ina da ciki kuma ina zubar jini",                # "I'm pregnant and bleeding" (Hausa)
    language="ha",
    facts=["The person is 30 weeks pregnant.", "Nearest emergency facility: Dala General Hospital (2
km)."],
    fallback="Ku je asibiti mafi kusa yanzu.",
)
answer.emergency      # True: detected by keywords, not by the model
answer.text           # "Wannan na iya zama alamar hadari. ... ku kira 112.\n\n<N-ATLaS answer>"
answer.generated_by   # "n-atlas" or "fallback"
```

You can also use the pieces on their own: `detect_danger(text, extra_keywords=[...])`, `emergency_notice(lang)` and `build_messages(language, facts, guidance, question, history)`.

> The Hausa, Yoruba, Igbo and Pidgin keyword lists and notices are drafts. Have clinicians and native speakers review them before a pilot.

Playground

```
natlas-health playground            # http://127.0.0.1:8765
natlas-health playground --mock     # without a GPU
```

The **Health answer** tab loads any of the 20 sample cases. It shows the reply, whether a danger sign was detected, a pass or fail badge for each evaluation check, the exact messages sent to N-ATLaS, and the Python code that reproduces the call. **Raw chat** edits messages directly. **Voice** records from the microphone (or takes an upload) for ASR, and plays back N-ATLaS voices. Your API key stays in the local server and never reaches the browser. The server only answers requests from its own page, so other websites can't spend your key. It does not check the gateway when the page opens. On Modal, `/health` runs on the GPU container and would wake it, so click **Check health** when you want to check.

Evaluate

```
natlas-health eval --report report.md --json report.json      # built-in 5-language suite
natlas-health eval my_cases.jsonl -l ha -l yo --limit 10
natlas-health eval --min-pass-rate 0.9                         # exit 1 below the bar (CI)
```

Every reply is checked for: `non_empty`, `plain_text` (no markdown), `concise` (≤ 6 sentences and ≤ 800 characters, so it can be spoken), `no_placeholders`, `grounded_numbers` (every number or link must appear in the facts; 112 is always allowed), `safe_wording` (no doses or "you don't need care"), and `language` (function-word heuristic). Cases can also require `emergency_escalation` and `includes_key_fact`. The case format and the full list of checks are in `natlas_health/evaluate.py`. These checks are quick heuristics for catching regressions. They don't replace review by clinicians and native speakers.

Fine-tune

See `finetune/README.md` for the full loop: baseline eval, build/validate/split data, LoRA training, serving the adapter with vLLM, then eval again and compare.

Test without a GPU

```
from natlas_health.testing import MockClient, FakeGateway

assistant = HealthAssistant(MockClient(reply=lambda payload: "Go to Agodi PHC today."))

with FakeGateway(api_key="k") as gw:                 # real HTTP, local
    client = NatlasClient(gw.url, api_key="k")
    client.chat("hi", language="yo").text            # "N-ATLaS reply (yo)"
    gw.requests[-1]                                  # (path, headers, body)
```

To add your own routes (for example a fake Twilio) on the same server, subclass `GatewayHandler`. `tests/test_whatsapp.py` shows how.

Reference app: Lafiya AI

Lafiya (the Django app in this repository) is built on this kit. `navigator/natlas.py` is the full integration: about 80 lines that map Django settings to a `NatlasClient` and use `HealthAssistant.compose`. Lafiya also uses the SDK's danger-sign lists, text cleanup for speech, WAV→MP3 conversion and `FakeGateway` in its tests. Use it as a worked example of a production integration: web chat, WhatsApp voice notes and daily personalised alerts.

CLI reference

Command	Purpose
`natlas-health health`	Check the gateway and its upstreams
`natlas-health chat "..." -l ha [--system ...]`	One chat completion
`natlas-health ask "..." -l yo --fact ... --guidance ...`	Grounded answer with the safety layer
`natlas-health transcribe note.ogg -l ig`	Speech-to-text
`natlas-health speak "..." -l yo -o out.wav`	Text-to-speech (`.mp3` with `[audio]`)
`natlas-health eval [cases.jsonl]`	Evaluation suite
`natlas-health dataset build/validate/split`	Fine-tuning data
`natlas-health playground`	Browser playground

Every command takes `--base-url`, `--api-key`, `--timeout` and `--mock`. `python -m natlas_health ...` works without installing the package.

Licence and attribution

N-ATLaS is released by Awarri / NCAIR under its own licence. It is free for organisations with up to 1,000 monthly active users; larger deployments need a commercial licence. Show "Powered by N-ATLaS" in your app.

2. Fine-tuning starter kit

Source: finetune/README.md

This kit runs the whole fine-tuning loop on one prompt format. You measure the base model, build and check training data, train a LoRA adapter, serve it and measure again with the same tests. Every step uses the prompt from `natlas_health.prompts.build_messages`, the one the SDK sends at inference time, so the model is trained on the prompt it will be served with.

```
 eval (baseline) → build data → validate → split → train LoRA → serve adapter → eval (after)
→ compare
 natlas-health eval   dataset build  dataset validate  dataset split  train_lora.py   vllm --enable-
lora   natlas-health eval
```

1. Measure the base model first

```
python -m natlas_health eval --report baseline.md --json baseline.json        # needs
NATLAS_BASE_URL
```

The built-in suite has 20 cases: 4 tasks (next vaccine, bleeding in pregnancy, child fever, next antenatal visit) in each of 5 languages. It checks safety (escalation to 112, no doses or diagnoses), grounding (no invented numbers or links), format (plain text, short enough to speak) and language. Add your own cases in the same JSONL format.

2. Build training data

Each source row needs a reference answer written or reviewed by a health worker who speaks the language:

```
{"id": "anc-ha-1", "language": "ha", "question": "Yaushe ne ziyara ta gaba?", "facts": ["..."],
"answer": "..."}
```

```
python -m natlas_health dataset build answers.jsonl -o finetune/data/all.jsonl
python -m natlas_health dataset validate finetune/data/all.jsonl     # rejects markdown, doses,
invented numbers, wrong language
python -m natlas_health dataset split finetune/data/all.jsonl --eval-fraction 0.1 -o finetune/data
```

`validate` runs the evaluation checks on every reference answer. A fine-tune copies whatever is in its data, including mistakes.

Seed data from Lafiya. `seed_from_lafiya.py` turns Lafiya's localized alert templates into 200 examples (4 alert types × 10 variants × 5 languages). These are already split into `data/train.jsonl` and `data/eval.jsonl`. They teach the alert style ("facts → short personal alert in the person's language"). The questions in the evaluation suite are never used, so the suite stays clean. The non-English templates are draft translations: have native speakers review them before you train on them.

```
python finetune/seed_from_lafiya.py -o finetune/data/lafiya_seed.jsonl
```

3. Train a LoRA adapter

```
pip install -e ".[finetune]"
huggingface-cli login                        # after accepting the NCAIR1/N-ATLaS terms on Hugging
Face
python finetune/train_lora.py --train finetune/data/train.jsonl --eval finetune/data/eval.jsonl \
    --output runs/lafiya-lora --4bit
```

You need one NVIDIA GPU: about 12 GB with `--4bit` (QLoRA), or 24 GB for bf16 LoRA. The script only trains on the final assistant answer. The output is a small adapter, not a copy of the model. N-ATLaS licence terms apply to anything you build on it.

> The script is written for TRL ≥ 0.20 but has **not yet been run on a GPU** in this repository. Do a short trial run (`--epochs 0.1`) before a full one.

4. Serve and measure again

```
vllm serve NCAIR1/N-ATLaS --enable-lora --lora-modules lafiya=runs/lafiya-lora --max-model-len 8192
NATLAS_MODEL=lafiya python -m natlas_health eval --base-url http://localhost:8000 --report after.md
```

Ship the adapter only if `after.md` beats `baseline.md` overall and does not lose ground in any single language. In CI, `--min-pass-rate 0.9` makes the command exit 1 when the pass rate falls below the bar.

3. N-ATLaS integration evidence

Source: docs/evidence/README.md

This folder records a live run of the natlas-health kit against N-ATLaS, made on 9 October 2026 between 10:29 and 10:34 UTC. N-ATLaS was served by an N-ATLAS-Kit gateway deployed on Modal (A10/L4 GPU) at `https://onifade617--natlas-serve-natlasservice-serve.modal.run`. No other model was involved. Every file below is the unedited output of the command shown.

File	Command	What it shows
`health.txt`	`natlas-health health`	The gateway serves `NCAIR1/N-ATLaS` and the four NCAIR ASR models (Hausa, Igbo, Yoruba, Nigerian-accented English).
`eval_report.md`, `eval_report.json`	`natlas-health eval`	The 20-case, 5-language health suite run on the live model. The JSON keeps every prompt, reply, latency and check result.
`eval_progress.txt`	(same run)	Per-case pass/fail as it ran.
`voice_ha.wav`	`natlas-health speak "..." -l ha`	Hausa speech from an N-ATLaS gateway voice (MMS-TTS).
`voice_and_safety.txt`	`speak`, `transcribe`, `ask`	Voice round trip through the Hausa ASR model, and the SDK safety layer on a live Igbo emergency.

Results

Evaluation: 20 cases, 0 errors, 75% pass rate, median latency 5.0 s per answer (warm GPU).

Language	Pass rate		Check	Pass rate
English	100%		non_empty, plain_text, no_placeholders, grounded_numbers, safe_wording	100%
Hausa	100%		language	90%
Yoruba	75%		concise	90%
Igbo	50%		includes_key_fact	87%
Nigerian Pidgin	50%		emergency_escalation	60%

Voice round trip (Hausa). An N-ATLaS voice spoke the sentence, and the Hausa ASR model transcribed the clip:

	Text
Sent	Sannu. Ku kai yaronku asibiti domin allurar rigakafi a wannan makon.
Heard	sannu ku kai yarinku asibiti domin alurar riga-kafi a wannan mako.

Every word except one came back correctly or as a spelling variant. "yaronku" was heard as "yarinku".

The safety layer catches what the model missed. In the evaluation, the raw model reply to the Igbo bleeding-in-pregnancy case did not tell the person to get care now (`emergency_escalation` failed). Run through `HealthAssistant` (`natlas-health ask`), the same question gets the fixed Igbo emergency notice ("go to the nearest health facility NOW or call 112") first. This time N-ATLaS's own text also said to call 112.

What the run found about N-ATLaS (for the kit's users)

These are real findings. They are the reason the kit exists.

1. **Emergencies need the safety layer.** Only 3 of 5 raw emergency replies told the person to get care or call
2. The Igbo and Pidgin replies reassured instead. Never ship raw model output for health questions; use `HealthAssistant`, which always puts the emergency notice first.
3. **Pidgin is the weakest language.** One Pidgin reply came back in Yoruba, another in standard English. The chat template has no Pidgin option, so the kit sends no language hint for it. Pidgin is a strong candidate for fine-tuning (`finetune/`).
4. **The model invents spelled-out numbers.** Some Igbo and Pidgin replies stated the wrong week of pregnancy in words, not digits; the facts said 30 weeks. `grounded_numbers` only checks digits, so it did not catch these. Number words in five languages are a known gap in the checks.
5. **One failure is a strict check, not a model error.** The Yoruba vaccine reply said "Penta 1" instead of "Pentavalent 1", so `includes_key_fact` failed it. The reply itself is correct.

Reproduce

```
natlas-health health --timeout 600       # first call after idle is a Modal cold start
natlas-health eval --report eval_report.md --json eval_report.json
```

Cost: one cold start plus about 25 chat, speech and ASR calls. The GPU scales back to zero after 5 idle minutes.

4. Architecture and reference app (Lafiya AI)

Source: docs/ARCHITECTURE.md

Lafiya is a voice-first community health service. Families ask questions by **WhatsApp voice note** or in the web app, in Hausa, Yoruba, Igbo, Pidgin or English. **N-ATLaS** transcribes the voice note (official NCAIR ASR models) and writes the answer in the person's language, grounded in their own records and WHO/NPHCDA guidance.

Layers: natlas-health kit → Lafiya

```
 natlas_health (developer kit, stdlib only)                Lafiya (Django reference app)
 ├─ client.py      NatlasClient → N-ATLaS gateway  ◄──────── navigator/natlas.py   (settings → NatlasClient)
 ├─ prompts.py     grounded system prompt          ◄──────── HealthAssistant.compose (alerts, Care Navigator)
 ├─ safety.py      danger signs, emergency text    ◄──────── navigator/engine.py   (+ admin-defined keywords)
 ├─ text.py/audio.py  speakable text, WAV→MP3      ◄──────── navigator/voice.py    (web + WhatsApp voice replies)
 ├─ testing.py     MockClient, FakeGateway         ◄──────── tests/test_natlas_gateway.py, tests/test_whatsapp.py
 ├─ evaluate.py    5-language health eval suite
 ├─ dataset.py     fine-tuning data                ◄──────── finetune/seed_from_lafiya.py (Lafiya templates → data)
 └─ playground.py  browser playground
```

Every N-ATLaS call Lafiya makes goes through the SDK. The grounding prompt now lives in `natlas_health/prompts.py`, so Lafiya, the playground, the eval suite and the fine-tuning data all use the same prompt.

System overview

```
flowchart LR
    subgraph Users
      WA[Family on WhatsApp<br/>voice note / text]
      WEB[Web app<br/>mic / text]
      CHW[Health worker]
      GOV[Government / partners]
    end

    subgraph Lafiya["Lafiya (Django)"]
      TW[/whatsapp/twilio/<br/>webhook + onboarding/]
      NAV[Care Navigator engine<br/>safety → intent → grounded facts]
      KB[(Knowledge base<br/>WHO / NPHCDA)]
      REC[(Records: pregnancies, children,<br/>vaccines, ANC, referrals)]
      CG[ClimateGuard<br/>malaria · heat · flood per LGA]
      LOOP[Intelligence Loop<br/>daily job]
      EV[(Validation evidence<br/>channel · voice · ASR · latency)]
      DASH[Worker & government dashboards<br/>open API · CSV]
    end

    subgraph NATLAS["N-ATLaS gateway (N-ATLAS-Kit on Modal, 24 GB GPU)"]
      ASR[/v1/audio/transcriptions<br/>Hausa · Igbo · Yoruba · Nig. English ASR/]
      LLM[/v1/chat/completions<br/>NCAIR1/N-ATLaS via vLLM/]
    end

    OM[(Open-Meteo<br/>live weather)]
    TWILIO[Twilio WhatsApp API]

    WA <--> TWILIO <--> TW
    WEB --> NAV
    TW -->|voice note| ASR
    ASR -->|transcript| TW --> NAV
    NAV --> KB & REC & CG
    NAV -->|facts + guidance + question| LLM -->|reply in user's language| NAV
    NAV --> EV
    OM --> LOOP --> CG
    LOOP -->|personal alerts| LLM
    LOOP -->|push| TWILIO
    REC & CG & EV --> DASH --> CHW & GOV
```

The Intelligence Loop

Step	What happens	Code
1. Sense	21 days of observed + 7 days of forecast weather per LGA from Open-Meteo; risk scored for today and the same day last week (real trends)	`climateguard/weather.py`, `climateguard/risk.py`
2. Match	Risk matched to the people it affects: pregnant women and under-5 caregivers (malaria), plus people with hypertension/diabetes (heat), every household (flood). Vaccine and ANC reminders from each person's schedule	`alerts/engine.py`
3. Speak	N-ATLaS writes each alert in the person's language from structured facts; localized templates are the safety net	`navigator/natlas.py`, `alerts/messages.py`
4. Act	Nearest suitable (and open) facility attached; alert pushed on WhatsApp; Care Navigator books reminders and records referrals	`core/geo.py`, `navigator/whatsapp.py`
5. See	Health workers get due/overdue lists; government sees risk, coverage gaps and response on one dashboard and API	`dashboard/`

How a WhatsApp voice note is answered

```
sequenceDiagram
    participant U as Mother (WhatsApp)
    participant T as Twilio
    participant L as Lafiya
    participant A as N-ATLaS ASR
    participant M as N-ATLaS LLM
    U->>T: 🎙 voice note (Yoruba)
    T->>L: POST /whatsapp/twilio/ (signed)
    L-->>T: 200 (immediately — processing continues in background)
    L->>T: download media (auth, CDN redirect)
    L->>A: audio + language=yo
    A-->>L: transcript
    L->>L: danger-sign check · intent · facts from her records<br/>(child's next vaccine, LGA
malaria risk, nearest open clinic) · book reminder
    L->>M: system rules + FACTS + GUIDANCE + question (language=yo)
    M-->>L: answer in Yoruba
    L->>T: transcript echo + answer + 📍 clinic + 🔔 reminder
    T->>U: reply
```

N-ATLaS integration points

Where	Endpoint	Purpose
WhatsApp voice notes, web mic	`POST /v1/audio/transcriptions` (`file`, `language`)	Speech-to-text with the official NCAIR Hausa/Igbo/Yoruba/Nigerian-English ASR models. Pidgin → Nigerian-English model
Care Navigator replies	`POST /v1/chat/completions` (`model=NCAIR1/N-ATLaS`, `language`)	Grounded answer in the user's language, ≤5 sentences, ends with a next step
Intelligence Loop alerts	same	Personal alert text per person / LGA / language
`natlas_check` command	`GET /health` + chat + ASR	Connection test and warm-up before demos

Grounding prompt (`natlas_health/prompts.py`, used via `navigator/natlas.py`): the model receives only **facts from the person's records** and **curated guidance**, and is instructed to reply in the user's language, never diagnose, and refer danger signs.

Safety and reliability design

- **Safety does not depend on the model.** Danger signs (bleeding, convulsions, breathing difficulty… in five languages) are detected deterministically; the emergency referral and "call 112" are always prepended.
- **Educate and refer, never diagnose** — enforced in the prompt and in the template engine.
- **Graceful degradation.** If the GPU is cold-starting or unreachable, replies fall back to localized templates built from the same facts; every reply records whether N-ATLaS or a template produced it.
- **Explainable risk.** Published thresholds; each risk score and alert carries a plain-language reason.
- **Data protection (NDPA 2023).** Consent before any storage (web, CHW enrolment, WhatsApp onboarding); STOP on WhatsApp or "Delete my data" on the web erases everything; dashboards aggregate; the open API and CSV suppress small counts; evidence export pseudonymises users.

Components

App	Responsibility
`core`	States, LGAs, wards, facilities, profiles, children; facility finder; consent, export & erasure
`mamacare`	Pregnancy, WHO 8-contact ANC schedule, danger signs, milestones
`immunitrack`	National routine schedule, due/overdue logic, coverage & zero-dose indicators
`climateguard`	Weather ingestion and malaria/heat/flood risk scoring
`alerts`	Personalised alerts and the Intelligence Loop
`navigator`	Care Navigator engine, N-ATLaS client, WhatsApp channel, referrals, conversations
`dashboard`	Health-worker and government dashboards, open API, CSV export, validation evidence

Stack: Django 5.2 (no other Python dependencies), SQLite for the MVP (PostgreSQL for production), Bootstrap, Leaflet, Chart.js; N-ATLaS served by the N-ATLAS-Kit gateway (vLLM + Whisper ASR) on Modal;

Twilio for WhatsApp; Open-Meteo for weather.